

OSCAR CHENG

COMPUTER SCIENCE STUDENT | SEEKING SOFTWARE INTERNSHIP

CONTACT

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[Personal Website](#)

[HackMD \(Tech Notes\)](#)

[Note Garden \(Quartz\)](#)

EDUCATION

B.S. in Computer Science

National Changhua University of
Education (NCUE)

September 2023 – June 2027 (Expected)

Changhua, Taiwan

SKILLS

Languages

Python, C++ (basics), JavaScript,
HTML/CSS

Backend & APIs

FastAPI, API integration

AI & Data

LLM API integration, RAG, vector search,
scikit-learn basics

Databases & Tools

Git/GitHub, Supabase, Qdrant, SQLite,
Linux basics

Core Areas

AI Agents, RAG, LLM Applications,
Backend Development

LANGUAGES

Mandarin Chinese — Native

English — Intermediate

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SUMMARY

Computer Science undergraduate at NCUE with hands-on experience building AI agents, Retrieval-Augmented Generation (RAG) systems, and Python-based backend applications. Experienced in designing end-to-end AI workflows involving LLM API integration, vector search, FastAPI, and external tool integration through multiple real-world projects. Passionate about building practical AI systems that are structured, reliable, and maintainable, with a strong interest in transforming language models into real-world applications.

PROJECTS

Market Agent — AI-Powered Market Research Assistant

Python, FastAPI, Supabase, OpenRouter, yfinance, scikit-learn • Jun 2026 – Present

Repo: <https://github.com/oscar940327/market-agent>

- Built a controlled AI agent that transforms natural-language market questions into structured research reports by coordinating price, technical, fundamental, news, and ML analysis modules.
- Developed backend workflows integrating LLMs, external APIs, Supabase, and automated data pipelines for collecting, updating, and organizing market data.
- Designed evidence-quality and freshness validation mechanisms to improve report reliability and clearly communicate the limitations of underlying data and AI-generated signals.
- Added machine learning reference signals for short-term upside probability and large-drop risk, together with evidence-quality and freshness checks to clearly communicate data reliability and model limitations.

RAG Financial Report Analysis

Python, Qdrant, Cross-Encoder, OpenAI API, Streamlit • Fall 2025

Repo: <https://github.com/oscar940327/RAG-Financial-Report-Analysis>

- Built a two-stage Retrieval-Augmented Generation pipeline to analyze 100+ page financial reports from Tesla and Palantir using reranking.
- Integrated a Cross-Encoder reranking model and standalone query generation to improve retrieval quality for multi-turn and cross-lingual questions.
- Reduced hallucination risk by constraining responses to retrieved evidence and requiring the model to explicitly state when the available context was insufficient.

VideoNote — AI-Powered Video Learning Note System

Python, FastAPI, OpenRouter, Whisper, Obsidian, Quartz • Jul 2026 – Present

Repo: github.com/oscar940327/video-note

- Built an end-to-end application that converts YouTube and Bilibili videos into editable, structured Markdown notes using available subtitles or Whisper transcription.
- Designed a multi-model LLM workflow for long-transcript processing, note generation, independent review, and automatic knowledge-base classification.
- Developed backend workflows for safely storing and reorganizing notes in an Obsidian Vault and publishing them as a knowledge website through Quartz and GitHub Pages.